Contamination Control Cleanroom Management

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Contamination Control Cleanroom Management (A09761)

Short Title: Contamination Control
Faculty Science and Computing
Department Science
Cluster Biology
Author BWHELAN
Credits 10 **Level:** Introductory

Description of Module / Aims

This module covers aspects of Cleanroom design, installation and maintenance. Students will be able to work in a Grade A/B Cleanroom and learn the importance of cleaning, routine operation and monitoring to maintain the integrity of the area to produce Parenteral products.

Programmes

`FDSC-0011`	Higher Certificate in Science in Good Manufacturing Practice and Technology (`WD_SGMPT_C`)	2 / 3 / M

Indicative Content

- Cleanroom design and classification; fundamentals of air filtration: principles of HEPA filtration and HVAC systems
- Disinfectant efficacy testing, testing the effectiveness of cleaning agents; cleaning validation
- Media fill trials using various combinations of equipment/staff/shift patterns; environmental monitoring programmes; particulate, microbial, in the 'at rest' and 'in operation' states
- Barrier/isolator technology as used in the control of contamination in pharmaceutical manufacturing facilities
- Clothing and housekeeping practices for the clean room staff and maintenance contractors; critical control of clean room entrance practices
- Principles of steam sterilisation and vaporised hydrogen peroxide
- Cleaning, decontamination and segregation of working areas within clean rooms; introduction to specialised methods of cleaning and decontaminating clean rooms

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Apply the principles of A/B Cleanroom design and associated barrier technology for sterile manufacturing.
2. Complete in-media simulation trials to qualify the cleanroom facility, personnel and operations.
3. Explain how to de-contaminate and monitor the A/B Cleanroom facility.
4. Describe a cleaning validation strategy.

Learning and Teaching Methods

- Lectures.
- Review of literature on aseptic processing.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,4
Continuous Assessment	50%	
In-Class Assessment	50%	2,3

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out practical/professional tasks.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/ professional tasks.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Essential Material

- "Parenteral Drug Association." https://www.pda.org/
- Midcalf, B., W.M. Philips, J.S. Neiger and T.J. Coles. _Pharmaceutical Isolators: a guide to their application, design and control_. London: Publications division of the Royal Society of Chemistry, 2004.
- Ramstorp, M. _Contamination Control in Practice_. Germany: John Wiley & Sons, 2008.
- Whyte, W. _Cleanroom Tec__hnology: fundamentals of Design, Testing and Operation_. 2nd ed. UK: John Wiley & Sons, 2010.

Supplementary Material

- "Irish Cleanroom Society." http://www.cleanrooms-ireland.ie/
- "Pharmaceutical Inspection Scheme." http://www.picscheme.org/